Elementary Statistics & Python

Rethinking How to Enhance MATH 1401

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November 4, 2026

Why Use Python in Elementary Statistics?

Why would one choose technology from 1999 when our students could benefit from the actual tools statisticians and data scientists use in real life?

1. Intuitively-named **modern tools** that are **simple to use** .
2. Authentic **large data sets** and the analysis tools for such.
3. **Flexible data structures** like tables, arrays, string variables, and matrices.
4. **Newer data types** and structures that go beyond a list of numbers.
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There is **no technology-free version** of Elementary Statistics, only poor-tech versions.

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- Extraneous topics that are not really needed and are not quite statistics content.
- Awkward capabilities for students to perform **Exploratory Data Analysis** .

Why a Redesign: Extraneous Topics

Given the data science approach that informed development of course materials, the following content is awkward for a statistics course:

1. **Trig functions**
2. **Logarithms**
3. Change of datatype functions like **int(), float(), and str()**
4. **Lists**
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These topics may be valuable, but they are neither in the current MATH 1401 syllabus nor taught by our colleagues in trad stats. While utilizing **Python** as the technology tool in statistics enabled by the **datascience**, **numpy**, and **scipy** packages may not be perfect, this setup provides an **immensely productive launching pad.**

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Taking notes from the datascience package and its easy-to-use, **intuitive functions** , I created six helper functions for exploratory data analysis.

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A **key feature** of these functions is that they will be **table-array agnostic** allowing students to use whatever format the data happen to be currently stored in.

Function Parameters for all the Functions

For ease of student use, all 3 functions use the same parameters in the same order (`show_means` is added for box plots):

```
box( data, num = None, sm = False)
```

The parameters work as follows:

- **data:** array_name or table_name
- **num:** column_title for numeric variable
- **sm:** the show_means setting

With `num=None`, the function anticipates array data yet allows table data when a column title input is specified.